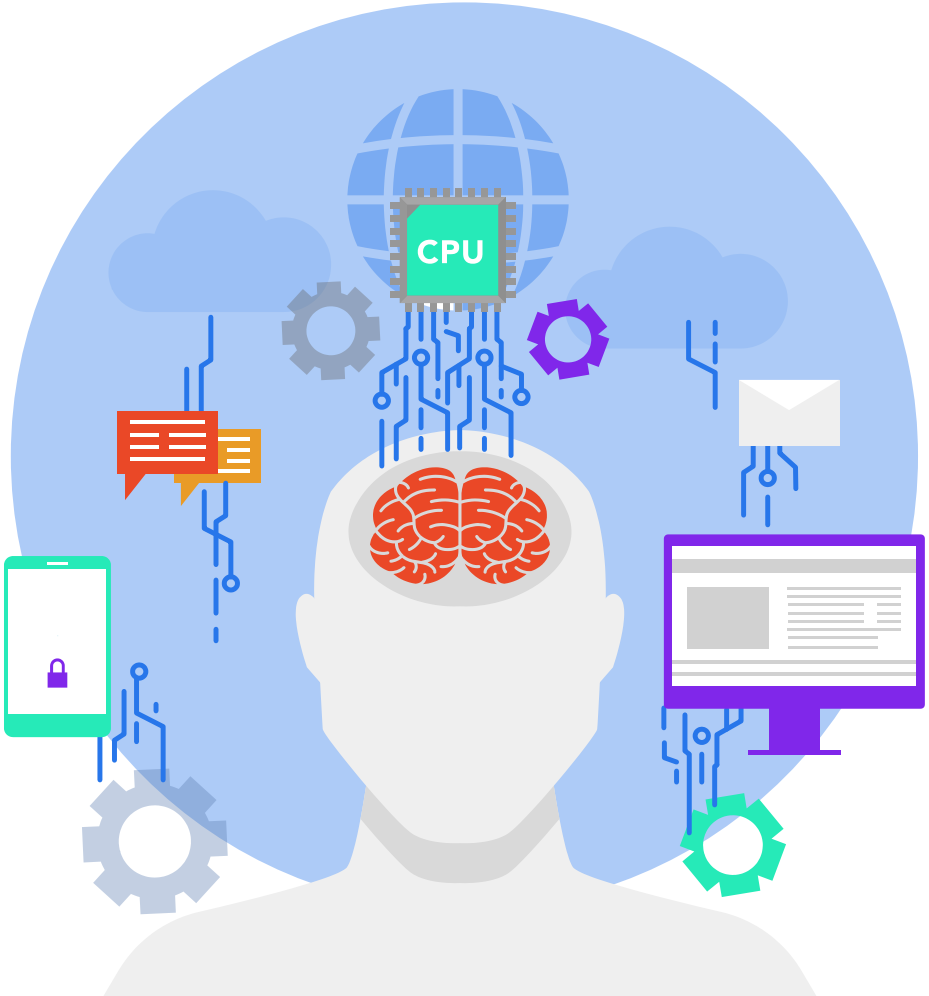
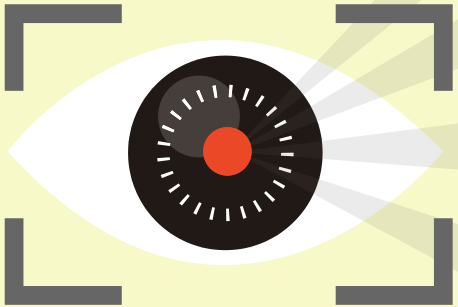


OnTime AI

Júlia Escoriza, Sandra de Olano, Oriol Pros & Sara Gibert

Problem

Unpredicted delay of public transportation



Common — Recurrent delays in big cities such as BCN

Unawareness — Users are often unaware of issues

Time Waist — Delays tend to make users waste unnecessary time

Target — Daily commuters in Barcelona, such as students

Dataset

Where we get it from?

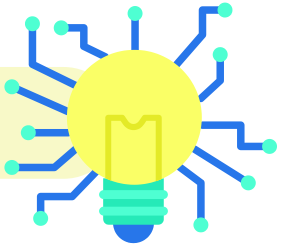
- **Transports Metropolitans de Barcelona** (possibly incidents or traffic conditions)
- **Open Data BCN** (schedules, real arrival times)

Type of Data

- Numerical variables (time or delay duration)
- Categorical variables (line, day of the week or month)

Used Data

What the user inputs vs what it gets as a result?



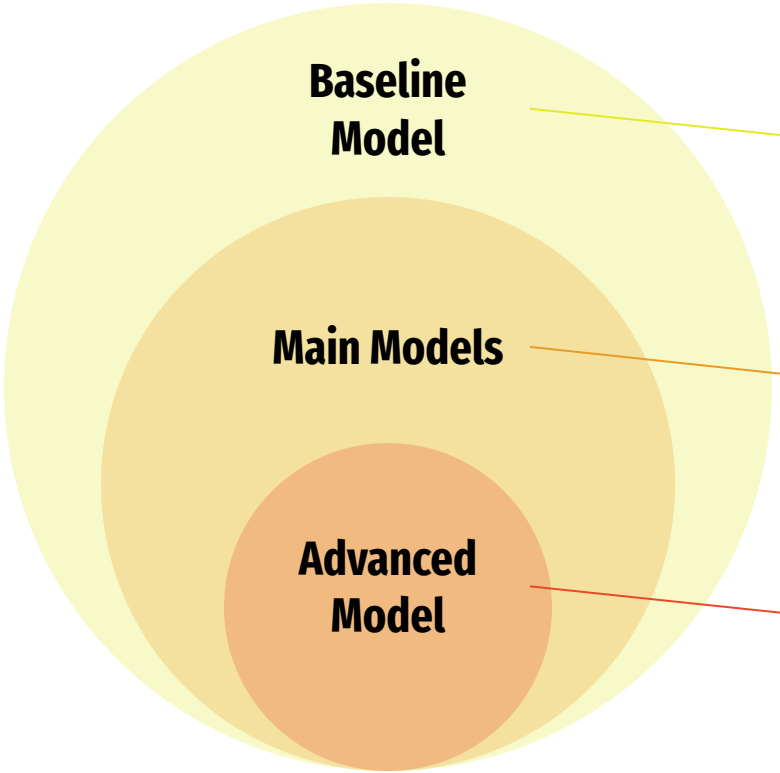
Input

- Time of day
- Day of week
- Line
- Stop
- Historical delays
- Traffic / weather (if time left)

Output

- Delay (minutes)
- Delay yes/no
- Alternative Routes (if time left)

Machine Learning Approach

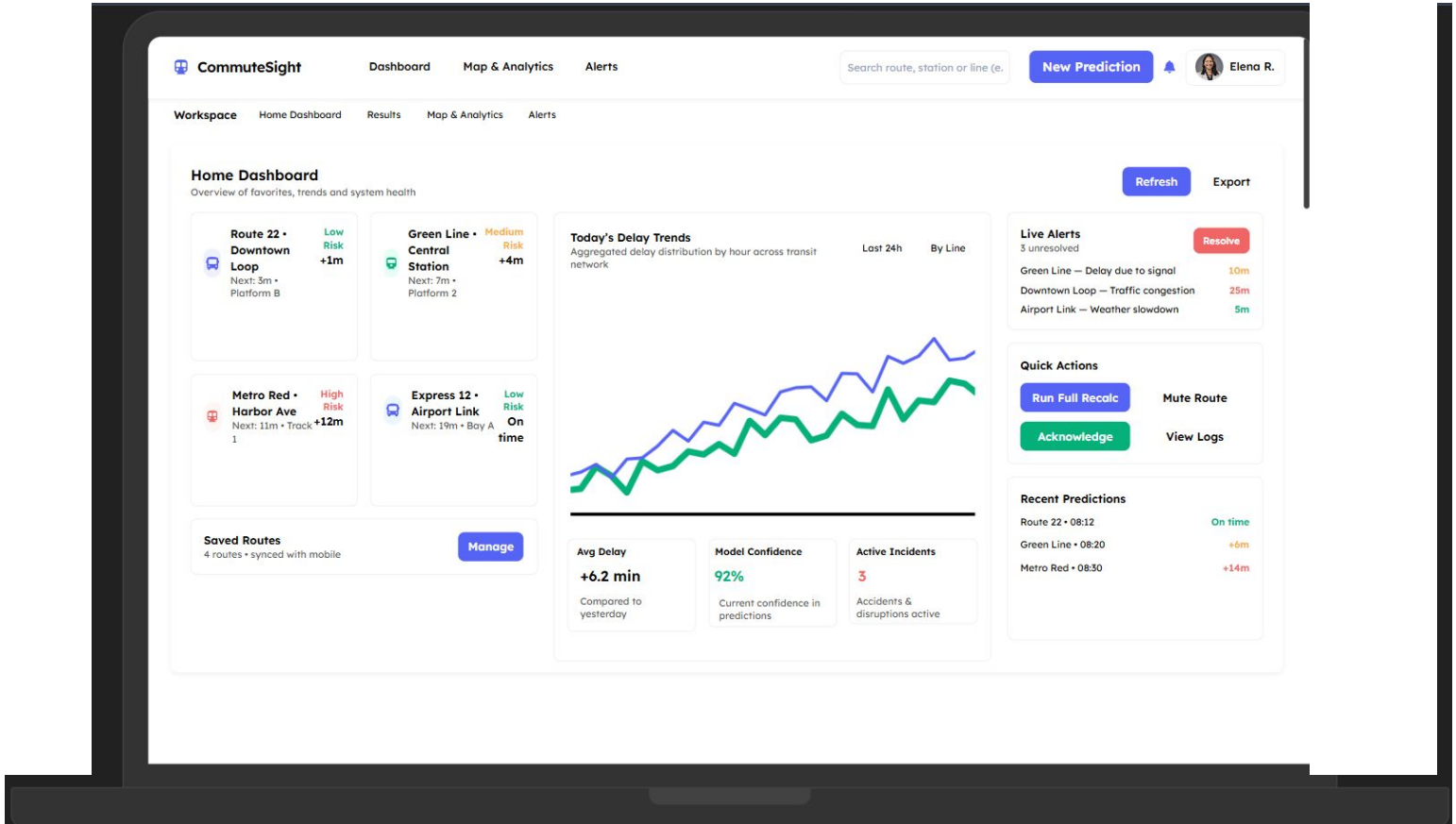


Predict delay: Linear Regression (minutes)
Logistic Regression (yes/no)

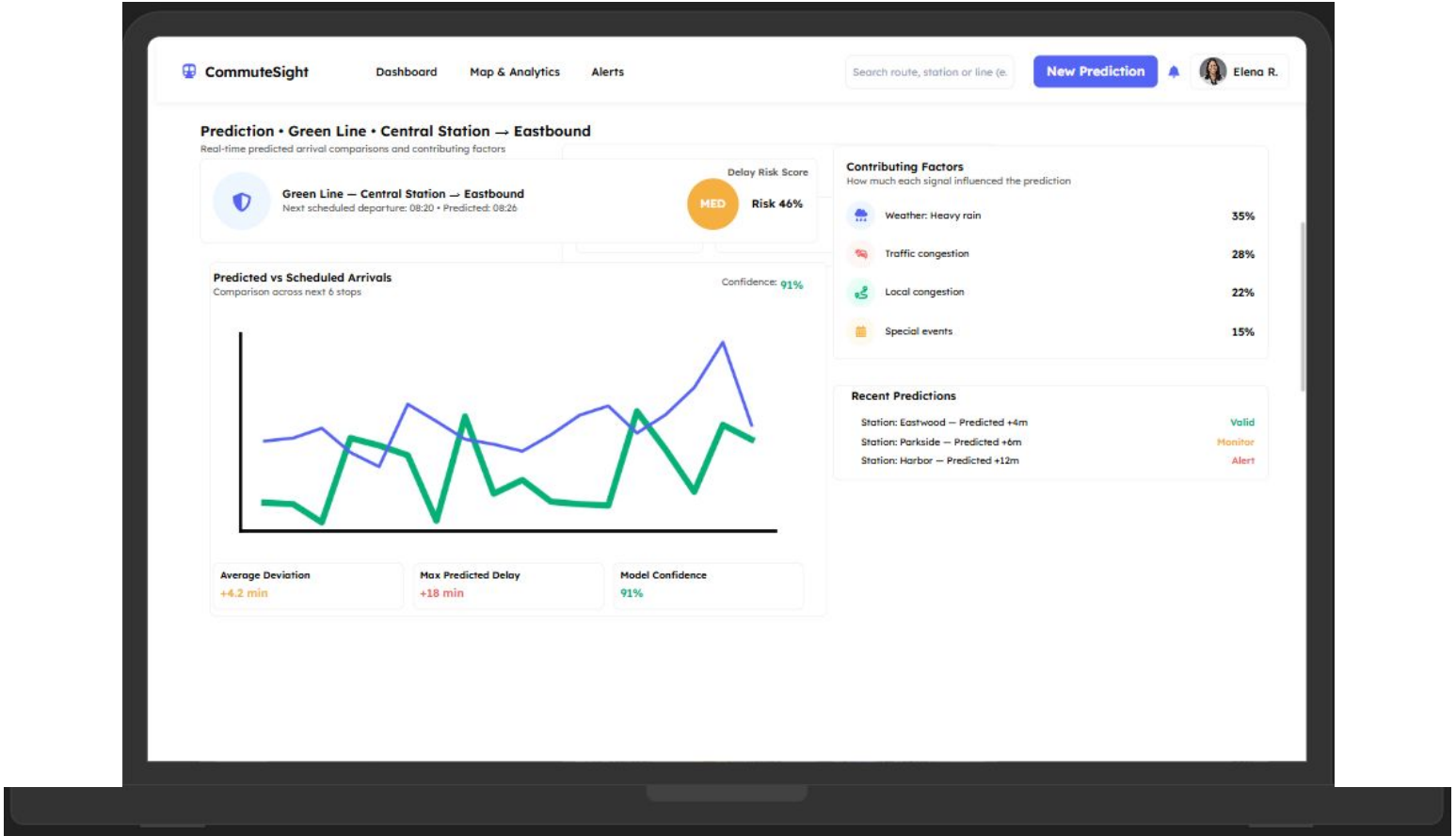
Decision Tree
Random Forest

Boosting & Time Series (if time left)

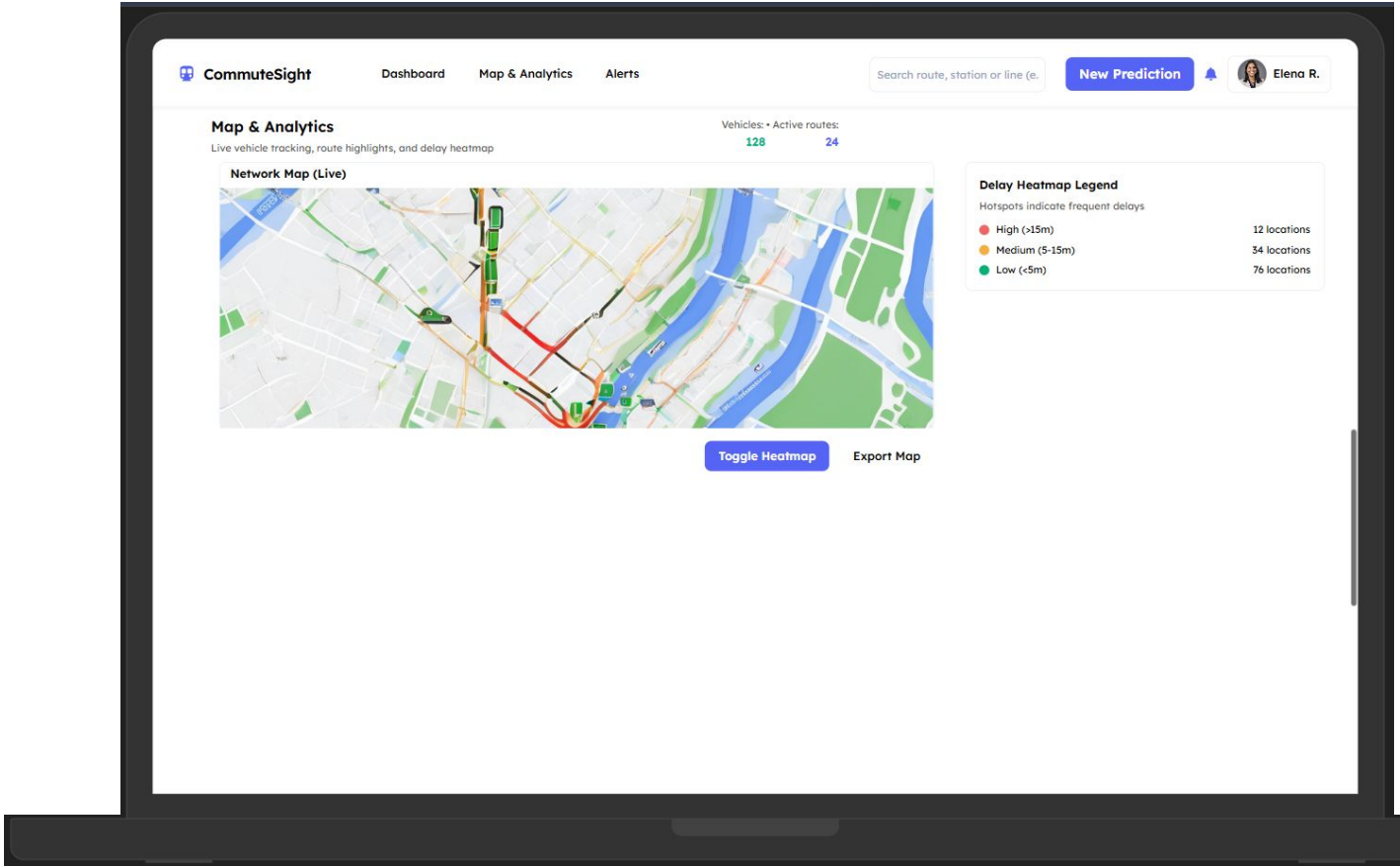
Mockups: Home Dashboard



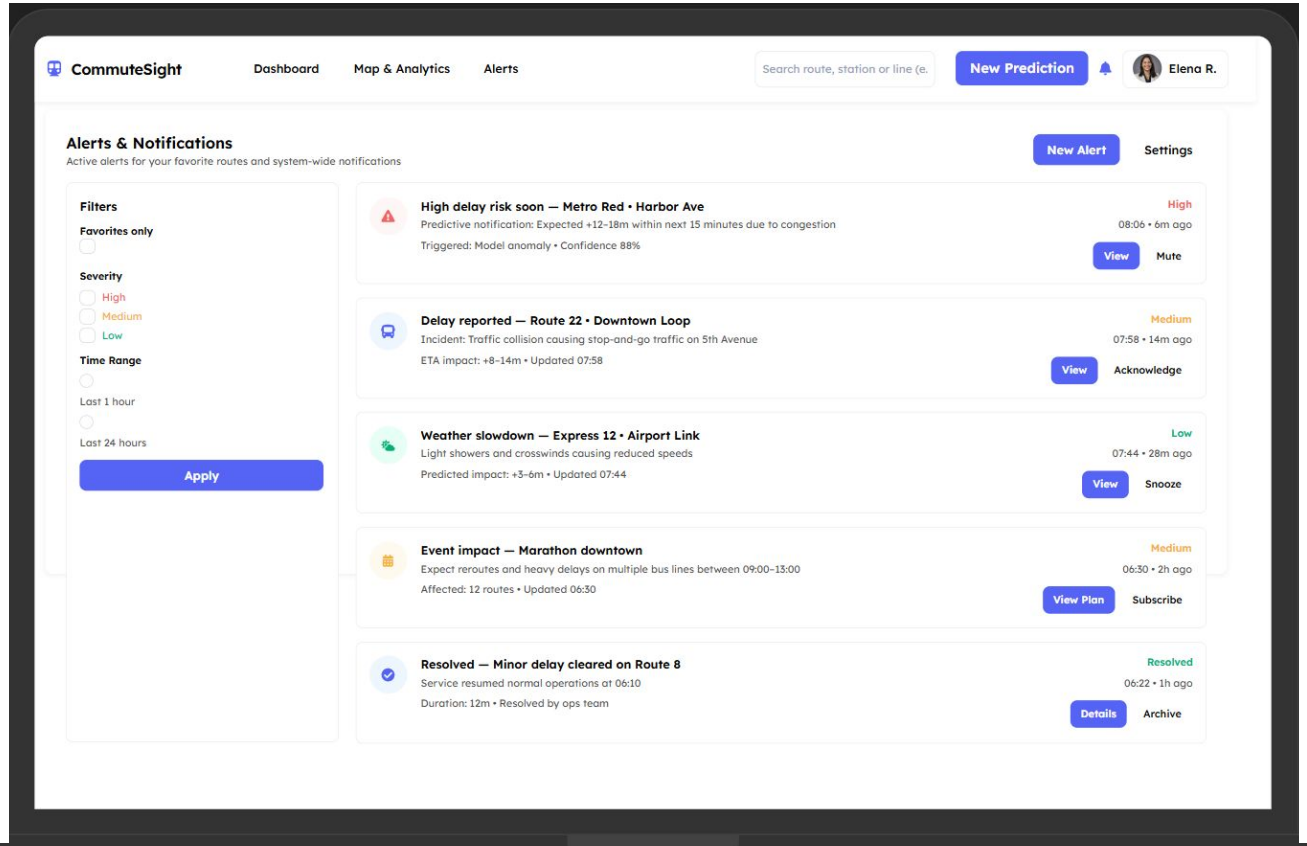
Mockups: Prediction



Mockups: Map & Analytics



Mockups: Alerts & Notifications



Thank you!